XAI Mid-Progress Report

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1. Project Overview

This project focuses on implementing high-performing machine learning models for the early prediction of cardiovascular diseases, based on methodologies from two key research papers:

1. **"Predictive Classifier for Cardiovascular Disease Based on Stacking Model Fusion"**.
2. **"Machine learning prediction in cardiovascular diseases: a meta-analysis"**.

2. Data Preprocessing and EDA

The dataset used is a processed cardiovascular disease dataset containing features such as Age, Gender, Cholesterol, and Blood Pressure.

- **Exploratory Data Analysis (EDA):** A correlation heatmap was generated to analyze the numeric relationships between features, ensuring no highly redundant variables were included initially.
- **Encoding:** Categorical variables were converted into numeric format using One-Hot Encoding to satisfy model requirements.
- **Linear Model Assumptions (VIF):** For the linear components of our models, the Variance Inflation Factor (VIF) was calculated to check for multicollinearity, ensuring the stability of the coefficients.
- **Feature Scaling:** Data was normalized using StandardScaler to bring all features to a common scale, which is essential for models like Logistic Regression and SVM.

3. Machine Learning Models Building & Results

I have implemented the primary "best" models identified in the research papers.

Model	**Source/Logic**	**Achieved Accuracy**	**Achieved AUC**
Logistic Regression (LR)	Base model and meta-learner used in Paper 1.	0.7318	0.7960

Stacking Ensemble	Proposed best model in Paper 1 using diverse learners.	0.7375	0.8062
Support Vector Machine	Identified as top-performing in Paper 2 meta-analysis.	0.7366	0.7958

- **Logistic Regression:** Used as a baseline and meta-learner with parameters C=1.0 and `solver='lbfgs'` to avoid overfitting.
- **Stacking Model Fusion:** Implemented using a first layer of Random Forest, Extra Trees, MLP, and CatBoost, with LR as the final meta-learner.
- **SVM:** Utilized an RBF kernel, which is highly effective for detecting non-linear patterns in clinical datasets.

4. Interpretability (XAI) Techniques and Results

Following the requirement to apply interpretability regardless of paper implementation, I applied four techniques to each model:

1. **SHAP (Global/Local):** Used to identify feature importance. In our stacking model, we specifically handled the `DimensionError` by creating a custom `Explanation` object to align SHAP values with the feature matrix.
2. **LIME:** Used to explain local predictions (e.g., explaining why Sample 0 was classified as "Disease").
3. **Permutation Importance:** Provided a global view of feature dependency by measuring how much accuracy dropped when features were shuffled.
4. **Partial Dependence Plots (PDP):** Visualized the marginal effect of risk factors like `AGE` and `CHOLESTEROL` on CVD probability, using `target=1` to focus on disease risk.

5. Challenges Faced

- **Target Identification:** The code was adjusted to dynamically identify the target column among several possible names (`CARDIO_DISEASE`, `HeartDisease`, etc.).
- **Multicollinearity Errors:** Fixed a `ValueError` in the VIF calculation where the length of values (12) did not match the index (11) after adding a constant.
- **XAI Reshape Errors:** Resolved a `ValueError` in PDP and an `AssertionError` in SHAP for the stacking model by ensuring class-specific indices and modern `Explanation` object formatting were used.

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6. Summary of Results and Comparative Analysis

Performance Comparison Table

Model	Implementation Accuracy	Implementation AUC	Paper Benchmark Accuracy	Paper Benchmark AUC
Logistic Regression (LR)	0.7318	0.7960	88.41%	0.94
Stacking Ensemble	**0.7375**	**0.8062**	**89.86%**	**0.95**
Support Vector Machine (SVM)	0.7366	0.7958	N/A*	0.92

Discussion of Results

The current implementation successfully demonstrates that the **Stacking Model Fusion** is the superior architecture, achieving the highest Accuracy (**0.7375**) and AUC (**0.8062**) among the three tested models. This aligns with the findings in Paper 1, which suggests that combining diverse learners (Random Forest, CatBoost, etc.) mitigates the bias of individual classifiers.

Analysis of the Performance Gap

There is a noticeable difference between the implementation results (~73.7%) and the paper benchmarks (~89.8%). This constitutes a primary "Challenge Faced" in the project, likely attributed to:

- **Dataset Variance:** The processed dataset (cardiovascular_diseases_processed.csv) contains different feature distributions and a larger sample size compared to the 918-sample Heart Dataset used in Paper 1.
- **Hyperparameter Optimization:** While the models are built, the specific "Optuna" automated tuning mentioned in the paper has not yet been fully applied to this specific dataset.
- **Feature Selection:** The original paper utilized SHAP-based feature selection to remove low-impact variables, a step currently earmarked for the next stage of development.